Optimizing Photometric Redshift Estimation Algorithms

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Abstract: With the imminent arrival of next-generation surveys like LSST, the demand for accurate photometric redshift (photo-*z*) estimates is higher than ever. We evaluated various existing photo-*z* estimation algorithms on a large simulated galaxy catalog in order to better understand and optimize their performance. This valuable information will soon be used to enhance algorithm performance, returning the most precise and reliable photo-*z* estimations.

Background:

Photo *z*s (Figure 1) are integral to galaxy evolution studies and cosmology, allowing the analysis of large datasets of objects whose redshifts are difficult to obtain due to the demands of spectroscopy. Photometric redshift estimation is a promising approach for tackling incoming large datasets; leading to the development of Redshift Assessment Infrastructure Layers[1] (RAIL), a framework for performing photo-*z*s. Many photo-*z* algorithms have been implemented in RAIL, though little work has yet been done to study and optimize the performance of these existing algorithms in the context of the LSST survey.

By testing these algorithms on a large simulated galaxy catalog and systematically adjusting their parameters, we aim to identify the most optimal configurations ("flavors") which yield the most accurate redshift estimates, ensuring these algorithms are good enough to meet the strict performance requirement needed to do precision cosmology with the vast datasets expected from next-generation surveys.

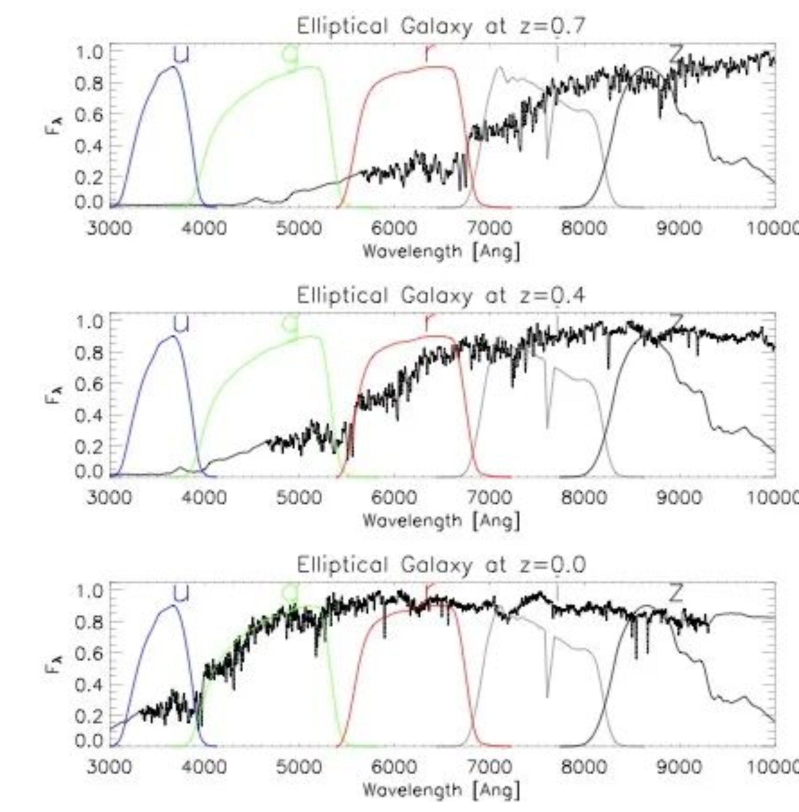

Figure 1: Flux is shown as a function of wavelength through spectral bands u, g, r, i, and z. As the galaxy increases in redshift, the observed spectrum shifts through the various photometric bands, changing the color ratios in each specific band

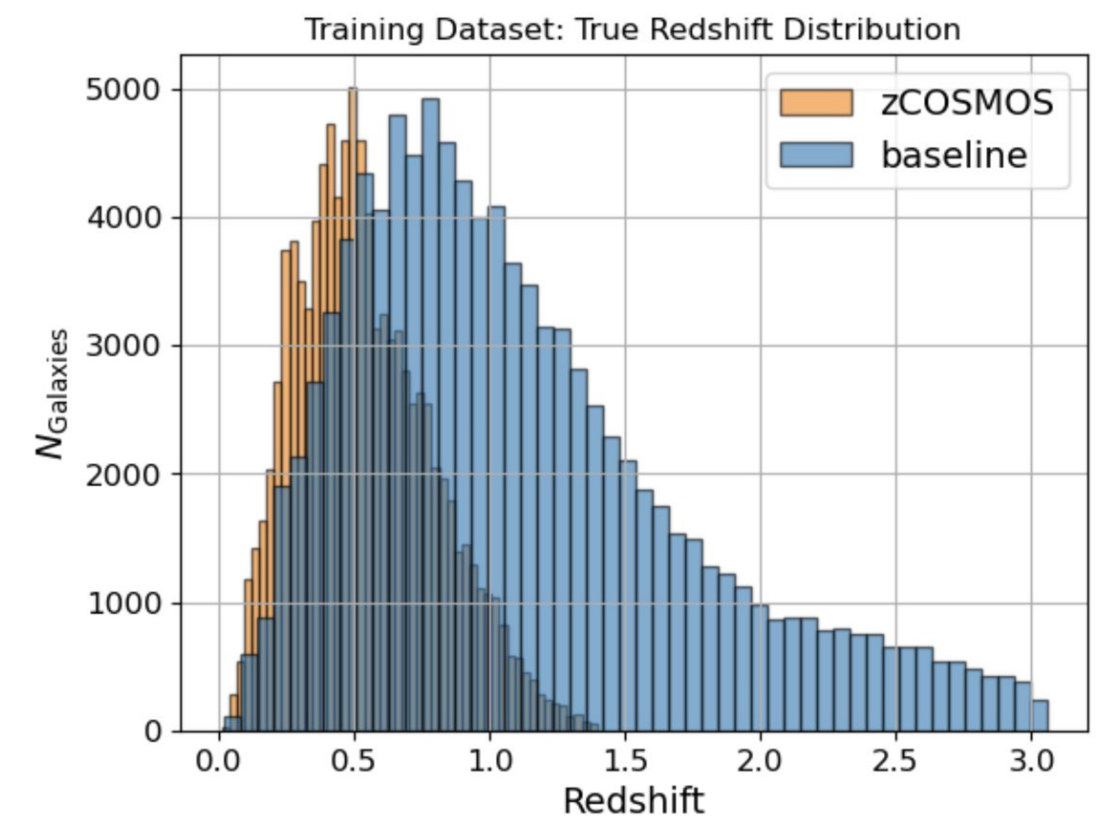

Figure 2: True redshift distribution of objects in representative and non-representative training sets.

Methodology:

We start with a simulated galaxy catalog of 5 million objects and first reduce the data by applying a magnitude cut (25.5 *i*-band mag), restricting the dataset to objects we are likely to use for cosmological analysis. Because these data do not include photometric errors, we apply errors based on a model of LSST[2] in order to replicate ground-based telescope conditions. Once this degraded sample is made, spectroscopic selections may be applied in order to create non-representative data which reflects the sort of sample we expect from spectroscopic surveys (Figure 2). From this sample, 100,000 objects are randomly selected. Various flavors (different configurations) of photo-*z* estimation algorithms are trained, on the representative and non-representative data, and tested. (See for example Table 1 and Figure 3). The estimated redshifts are evaluated through several metrics as well as tomographic binning evaluation.

Baseline

Algorithm	MAD
KNN[3]	0.021712
FZBoost[4]	0.016902
BPZ[5]	0.075951
GPZ[6]	0.024911
TPZ[7]	0.021164
SimpleNN[8]	0.031043

HSC

Algorithm	MAD
KNN	0.025237
FZBoost	0.018748
BPZ	0.075951
GPZ	0.042821
TPZ	0.030270
SimpleNN	0.130343

zCOSMOS

Algorithm	MAD
KNN	0.069136
FZBoost	0.052234
BPZ	0.075951
GPZ	0.161383
TPZ	0.218153
SimpleNN	0.151451

Table 1: Estimation results' median absolute deviation (MAD) is shown for each algorithm's default configuration using different spectroscopically selected training samples; HSC and zCOSMOS.

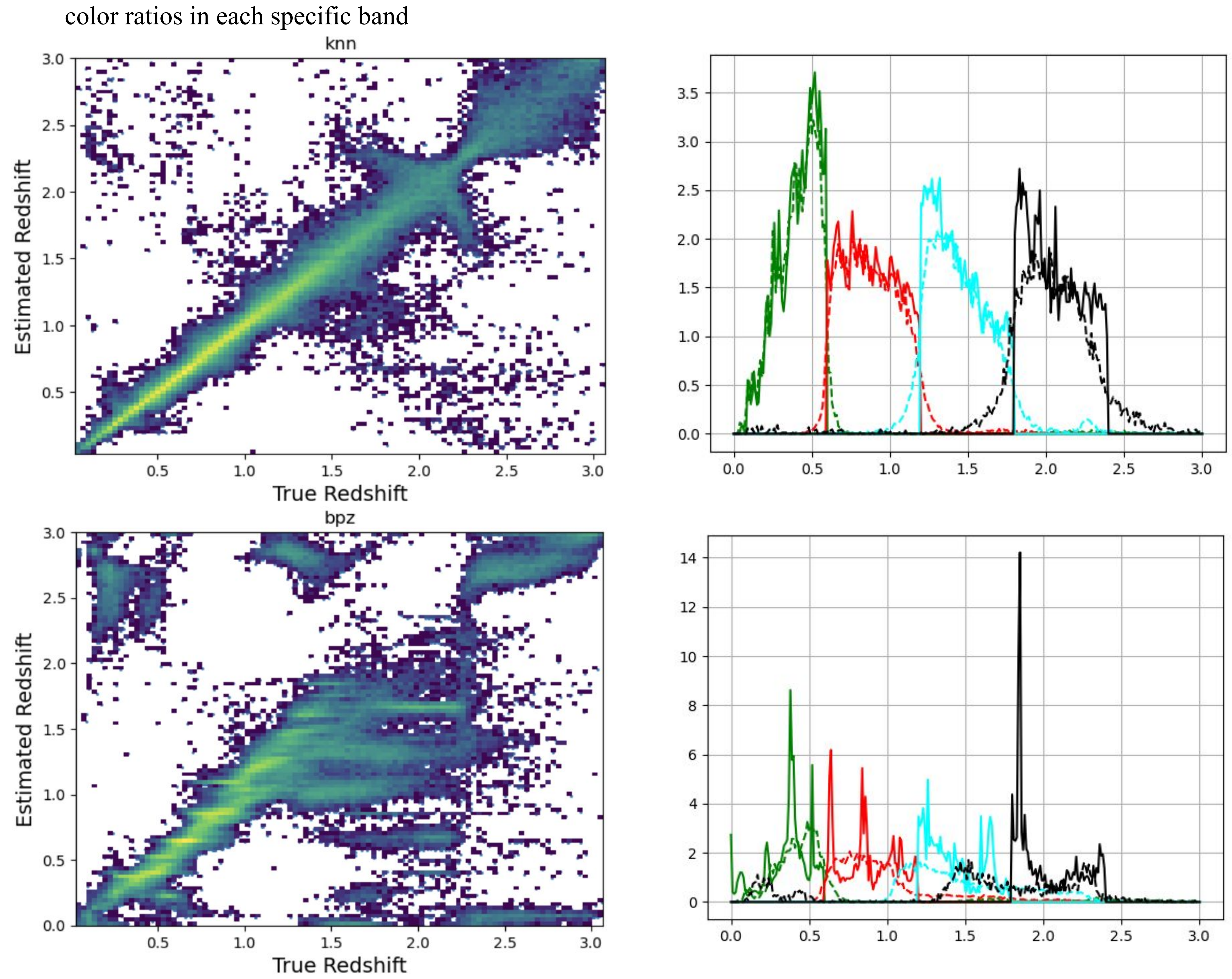

Figure 3: Photo-*z* estimation results for KNN and BPZ algorithms. The left column displays a 2D histogram showing the relationship between estimated and true redshift. KNN is a consistent top performer, with most of the estimates lying along the diagonal, while BPZ outputs less precise estimations. The right column presents tomographic binning plots, with true redshift values represented by the dashed line and estimated redshifts with a solid line. KNN's estimations track the true redshift values closer and with more consistency compared to other algorithms such as BPZ.

Conclusions and Future Work:

Upwards of 70 algorithm configurations have been tested so far, of which the results we have collected provide a better understanding of how these photo-*z* algorithms perform in this context. Initial results suggest KNN is a top performer as well as FZBoost, with the latter showing little degradation when training on a non-representative dataset. Next steps are to continue to explore and evaluate new configurations of the photo-*z* algorithms in order to determine how to retrieve the most accurate redshift estimations.

References:

[1] Code at https://github.com/lsstdesc/rail
[2] Code at https://github.com/jfcrenshaw/photerr
[3] Code at https://github.com/LSSTDESC/rail_sklearn
[4] Code at https://github.com/LSSTDESC/rail_flexzboost, algorithm described in Izbicki & Lee (2017), Dalmasso et al (2020)
[5] Code at https://github.com/lsstdesc/rail_bpz, algorithm described in Benitez (2000), Coe et al (2006)
[6] Code at https://github.com/LSSTDESC/rail_gpz_v1, algorithm described in Almosallam et al (2016)
[7] Code at https://github.com/lsstdesc/rail_tpz, algorithm described in Carrasco Kind & Brunner (2013)
[8] Code at https://github.com/LSSTDESC/rail_attic, algorithm described in Lee et al (2019)

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